

Question Paper

Exam Date & Time: 22-Dec-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
THIRD SEMESTER B.TECH. DEGREE MAKEUP EXAMINATIONS - DECEMBER 2025
DIGITAL SYSTEM DESIGN [CSE_2123]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Design a sequential circuit with two *JK* flip-flops *A* and *B* and two inputs *E* and *F*. If *E* = 0, the circuit (5)
A) remains in the same state regardless of the value of *F*. When *E* = 1 and *F* = 1, the circuit goes through the state transitions from 00 to 01, to 10, to 11, back to 00, and repeats. When *E* = 1 and *F* = 0, the circuit goes through the state transitions from 00 to 11, to 10, to 01, back to 00, and repeats.

Course Outcome: "CO4" / Cognitive Level: "Application"

- B) Show the transistor-level diagram for a two-input CMOS NAND gate. Determine why the NMOS (3)
transistors are placed in series in the pull-down network, while the PMOS transistors are placed in parallel in the pull-up network to achieve the NAND logic function.

Course Outcome: "CO1" / Cognitive Level: "Application"

- C) Simplify the function $F(a,b,c,d) = \text{IIM}(1,3,5,8,9,11,15) + \text{D}(2,13)$ using K-map to obtain the minimum (2)
cost SOP expression and draw the circuit.

Course Outcome: "CO1" / Cognitive Level: "Analysis"

- 2) Apply the concept of the Carry-Lookahead Adder (CLA) by deriving the general expressions for the (5)
A) Carry Generate (G_i), Carry Propagate (P_i), and the output carry (C_{i+1}) for a single stage. Apply these concepts to write the specific logical expressions for the output carries C_1 and C_2 in terms of the initial carry C_0 and the corresponding *G* and *P* terms. Explain why the CLA is faster than a Ripple Carry Adder.

Course Outcome: "CO2" / Cognitive Level: "Application"

- B) Design a 4-to-1 multiplexer using only 2-to-1 multiplexers. Implement this design in Verilog HDL (3)
and simulate the circuit.

Course Outcome: "CO2" / Cognitive Level: "Application"

- C) Illustrate a circuit using full adders to find the no. of 1's in a three-bit unsigned number. (2)

Course Outcome: "CO2" / Cognitive Level: "Application"

- 3) Illustrate the synchronous counter with the following repeated binary sequence: 0, 1, 2, 3, 4, 5, 6. (5)
A) Use JK flip-flops. Check if it is self-correcting or not.

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- B) Apply the necessary external logic connections to a JK Flip-Flop to convert it into an equivalent T Flip-Flop. Explain the simple modification required on the J and K inputs to achieve the toggle function when needed. (3)

Course Outcome: "CO3" / Cognitive Level: "Application"

- C) Apply the understanding of sequential circuits to design a basic RS latch using NOR gates. Demonstrate its working with a detailed circuit diagram and truth table. (2)

Course Outcome: "CO3" / Cognitive Level: "Application"

- 4) Construct an ASM chart for the pseudo code of a bit counter and explain it. (5)

- A) Course Outcome: "CO5" / Cognitive Level: "Application"

- B) Determine the operation of a Serial-In, Parallel-Out (SIPO) Shift Register. Explain the specific steps (in terms of clock cycles) required to load an 8-bit data word into the register and how the data is then accessed at the output terminals. (3)

Course Outcome: "CO4" / Cognitive Level: "Application"

- C) Apply the design principles of an Asynchronous (Ripple) Counter and the operational constraints of a Positive Edge-Triggered T Flip-Flop to implement a 4-bit UP counter in Verilog. (2)

Course Outcome: "CO4" / Cognitive Level: "Application"

- 5) Draw the standard symbol and the truth table for a Tri-state Driver. Apply the Tri-state Driver to explain how the outputs of two registers R1 and R2 are safely connected to a common bus line with the help of a diagram. (5)
- A)

Course Outcome: "CO5" / Cognitive Level: "Evaluate"

- B) Design a three-bit counter that counts in the sequence 0, 1, 2, 4, 5, 6, 0. (3)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- C) Show the relationship between the number of states N generated by a simple ring counter and the number of flip-flops k used. (2)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

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Event Name	Subject
058-MIT BLR-MAKEUP EXAM-2025-2026	CSE_2123:DIGITAL SYSTEM DESIGN

#	Course Outcome	Description	Marks
1	1	CO 1	5
2	2	CO 2	10
3	3	CO 3	5
4	4	CO 4	20
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
058-MIT BLR-MAKEUP EXAM-2025-2026	CSE_2123:DIGITAL SYSTEM DESIGN

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	10	33(66%)
4	Analysis	4	12(24%)
5	Create	1	5(10%)
6	Evaluate	0	0(0%)
	Total	15	50 (100%)